Manpreet Singh

AI Engineer

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Address: Halifax, NS, Canada

Education

Master of Computer Science
Jan 2023 – Pursuing

Dalhousie University – *Halifax, NS, Canada*

Core courses – Machine Learning, Deep Learning, Natural Language Processing and ML for Software Engineering Applications.

GPA (Graduate courses) – 3.9/4.3

Bachelor of Computer Applications
May 2016 – May 2019

Guru Nanak Dev University – *India*

Core courses – Programming languages, Software development, Data structures and algorithms, Computer architecture and Operating systems.

GPA – 3.2/4.0

Research Experience

Research Assistant
May 2023 – Current

Dalhousie University – *Halifax, NS, Canada*

- **Decoding the Black Box: Enhancing Explainability in Transformer Models (Thesis)**
  - Conducted in-depth research on improving the **interpretability** of Large Language Models (**LLMs**) under various **quantization** settings.
  - Examined the impact on neuronal activity, correlation, and entropy.
  - Employed the **Integrated Gradients** technique to pinpoint key neurons driving model predictions.
  - Extended previous research by **prompting** an open-source LLM to generate human-readable explanations of patterns learned by individual neurons.
  - Validated these explanations through pattern simulation with a separate LLM.
- **Unveiling Patterns and Biases: Manifold Learning in NLP Embeddings -** Evaluated **manifold learning** techniques for dimensionality reduction of **NLP token embeddings**, visualizing results to identify model-learned biases and assess algorithm performance in capturing linguistic nuances.

Work Experience

AI Solutions Engineer
Jul 2019 – Oct 2022

Tata Consultancy Services – *Chennai, India*

- Developed a GPT-3 powered bot designed to translate **English text into SQL queries**, featuring dynamic prompt engineering for optimized database interactions. The bot was specially tailored to interface with a **custom database schema**, providing a highly user-friendly and context-aware querying experience.

- Built an **NLP chatbot using Rasa** to streamline coding tasks, resulting in a 25% increase in developer productivity.
- Engineered a Computer Vision application that transforms **2D architectural blueprints into 3D** visual models, revolutionizing how architects and clients interact with plans.
- Conducted lectures on Python and Artificial Intelligence, **mentored new joiners**, and **supervised project** timelines and deliverables as part of the onboarding process for over **100** junior engineers.
- Excelled in an initial 20-week intensive training program focused on AI and machine learning, securing a rank within the **top 0.5%** among a cohort of 1000 trainees. This strong foundation contributed to my subsequent contributions in machine learning and AI projects, including chatbot and Computer Vision application development.

Recognition and Awards

- Recognized for AI innovation by presenting a cutting-edge Neural Style Transfer project to India's Honorable Education Minister during a high-profile visit to TCS Ignite.
- Secured first place in the Plasma Quiz L.K. college, Jalandhar, showcasing extensive knowledge in science and technology.

Skills

- **Programming Languages:** Python, JavaScript, Java, C, and SQL.
- **Machine Learning & AI:** Pytorch, Supervised Learning, Unsupervised Learning, Deep Learning, Neural Network Interpretation, NLP, NLU, Explainable AI, Transformers, LLMs, Chatbot Development, Prompt Engineering, Computer Vision, GANs, CNN, TensorFlow, Keras, Scikit-learn, and Rasa.
- **Data Analysis:** Statistics, Pandas, Numpy, Matplotlib, Seaborn, and EDA.
- **Software Development & DevOps:** Agile, Git, Docker, RESTful APIs, Flask, and Django.
- **Management & Leadership:**, Team Lead, Mentoring and Project Management.